

May – June 2026

Nuclear #1 & #2 — Missed-Chapter Catch-Up Plan

OWNER	DURATION	SCHEDULE	DEADLINE
Parker	8 weeks	Mon/Tue/Wed	Jun 30, 2026

Weekly Cadence

MON	TUE	WED	THU – SUN
4 hrs reading	4 hrs reading	2 hrs problems	off / buffer

Daily Reading Workflow (Mon & Tue)

Each 4-hour reading day runs as a single 3-stage cycle over that day's assigned chapters. Don't sequence chapter-by-chapter — batch them through the pipeline:

STAGE 1	STAGE 2	STAGE 3
1	2	3
Skim + note questions 1 – 2 hrs Fast pass through day's chapters; jot down concepts you don't grasp as questions.	AI LaTeX notes 30 min – 1 hr Feed your skim notes to an AI; get a structured LaTeX summary to study from.	Reread + answer ~1 hr Second pass targeting your own questions; write answers in the margins.

Total active time: ~2.5–4 hrs. Remaining time in the 4-hr block is buffer. This workflow lets you cover heavier page loads (Wk 1, Wk 2, Wk 7) without falling behind — the AI LaTeX stage is doing the heavy lifting of distillation for you.

Reading Sources

Short	Full reference	Chapters
Loveland	Loveland, Morrissey & Seaborg — Modern Nuclear Chemistry (Wiley, 2006 / 2nd ed. 2017)	8, 9, 10, 11, 12, 14, 18
Mann	Robert Mann — An Introduction to Particle Physics and the Standard Model (CRC, 2010)	3–6, 9–14, 19–25
LB	Lancaster & Blundell — Quantum Field Theory for the Gifted Amateur (Oxford, 2014)	7, 12–13, 16–20, 26, 46–47
Thomson	Mark Thomson — Modern Particle Physics (Cambridge, 2013)	13, 17 (June companion)

Two-Month Overview

Every week: read ~60–100 pp across Mon + Tue, then do ~4–6 problems on Wed. Problem count is held modest on purpose because Wed is only 2 hours — better to do a few problems thoroughly than skim many.

	Dates	Theme	Reading (Mon+Tue)	Wed Qs
Wk 1	May 4 – 6	Nuclear decay & reactions	Loveland 8, 9, 10, 11	5
Wk 2	May 11 – 13	Nuclear applications	Loveland 12, 14, 18	4
Wk 3	May 18 – 20	Particle-physics foundations	Mann 3, 4, 5, 6, 9, 10	6
Wk 4	May 25 – 27	QFT: propagators & S-matrix	LB 7, 12, 13, 16, 17, 18, 19, 20	6
Wk 5	Jun 1 – 3	Gauge invariance & QED	Mann 11, 12, 13, 14	4
Wk 6	Jun 8 – 10	Weak interactions	Mann 19, 20, 21 + LB 26	4
Wk 7	Jun 15 – 17	★ Higgs mechanism & EW breaking	Mann 22, 23 + LB 46, 47 + Thomson 17	5
Wk 8	Jun 22 – 24	★ Neutrino oscillations & capstone	Mann 24, 25 + Thomson 13, 17.8	5

■ Gold = Week 7 (Higgs). ■ Green = Week 8 (neutrino oscillations).

Reading the weekly pages

MON	TUE	WED
Reading — ~50% of week	Reading — ~50% of week	Problem session — tackle 4–6 problems in one focused block

MAY — Nuclear Foundations

Rebuild the missed Nuc #1/#2 material, from concrete nuclear chemistry (Loveland) through particle-physics foundations (Mann) to first-pass QFT (LB).

WK 1

Nuclear Decay & Reactions

May 4 – May 10 • 10 hrs

Goal: rebuild intuition for the three decay modes and how nuclear reactions are parameterized. ~126 pp of reading — the heaviest week. Read aggressively on Mon & Tue; Wed hits the essential problems.

MON	TUE	WED
Mon May 4 • 4h • reading	Tue May 5 • 4h • reading	Wed May 6 • 2h • problems
Loveland Ch. 8 Beta (β) Decay <i>21 pp • ~1.5h</i> Loveland Ch. 9 Gamma (γ)-Ray Decay [skim] <i>27 pp • ~1.5h</i>	Loveland Ch. 10 Nuclear Reactions <i>48 pp • ~2.5h</i> Loveland Ch. 11 Fission <i>30 pp • ~1.5h</i>	Loveland Ch. 8 Problems 3, 5 Loveland Ch. 9 Problem 2 Loveland Ch. 10 Problem 3 Loveland Ch. 11 Problem 4

END-OF-WEEK CHECK (WED NIGHT OR THU)

Sketch the β -decay spectral shape, write $\sigma(E)$ for Rutherford, and explain asymmetric fission-fragment mass distributions.

WK 2

Nuclear Applications

May 11 – May 17 • 10 hrs

Goal: close out Loveland with the applied/experimental chapters. ~116 pp. Ch. 14 and 18 are long but descriptive — lock in vocabulary on Mon/Tue.

MON	TUE	WED
Mon May 11 • 4h • reading	Tue May 12 • 4h • reading	Wed May 13 • 2h • problems
Loveland Ch. 12 Nuclear Astrophysics <i>32 pp • ~2h</i>	Loveland Ch. 18 Radiation Detectors <i>40 pp • ~3h</i>	Loveland Ch. 12 Problem 3
Loveland Ch. 14 Reactors & Accelerators [skim] <i>44 pp • ~1.5h</i>	Buffer / AI latex note review <i>— pp • ~1h</i>	Loveland Ch. 14 Problem 3
		Loveland Ch. 18 Problems 2, 4

END-OF-WEEK CHECK

Pick the right detector for a given particle/energy; describe a thermal-reactor fuel cycle.

WK 3

Particle-Physics Foundations (Mann)

May 18 – May 24 • 10 hrs

Goal: Nuc #2 territory begins. Six short Mann chapters (~96 pp). Formalism-heavy — this is the math foundation everything else rests on.

MON	TUE	WED
Mon May 18 • 4h • reading	Tue May 19 • 4h • reading	Wed May 20 • 2h • problems
Mann Ch. 3 Symmetries <i>17 pp • ~1.5h</i> Mann Ch. 4 Conservation Laws <i>12 pp • ~1h</i> Mann Ch. 5 Particle Classification <i>15 pp • ~1.5h</i>	Mann Ch. 6 Discrete Symmetries <i>16 pp • ~1.5h</i> Mann Ch. 9 Scattering <i>16 pp • ~1.25h</i> Mann Ch. 10 A Toy Theory <i>20 pp • ~1.25h</i>	Mann Ch. 3 Q1 Mann Ch. 4 Q2 Mann Ch. 5 Q2 Mann Ch. 6 Q3 Mann Ch. 9 Q2 Mann Ch. 10 Q3

END-OF-WEEK CHECK

Derive Noether in one line; list CPT assumptions; draw a tree diagram in Mann's toy theory.

WK 4

QFT: Propagators, S-matrix, Scattering (LB)

May 25 – May 31 • 10 hrs

Goal: eight short LB chapters (~66 pp). Each introduces a new machine. Take it slow on the propagator and S-matrix chapters — those are the load-bearing ideas.

MON	TUE	WED
Mon May 25 • 4h • reading	Tue May 26 • 4h • reading	Wed May 27 • 2h • problems
LB Ch. 7 Example Lagrangians <i>6 pp • ~0.75h</i>	LB Ch. 17 Propagators & Fields <i>10 pp • ~1.25h</i>	LB Ch. 17 Exercise (17.1)
LB Ch. 12 Canonical Quantization Ex. <i>8 pp • ~1h</i>	LB Ch. 18 The S-matrix <i>10 pp • ~1.25h</i>	LB Ch. 18 Exercise (18.1)
LB Ch. 13 Many Components / Massive EM <i>9 pp • ~1h</i>	LB Ch. 19 Feynman Diagrams <i>12 pp • ~1h</i>	LB Ch. 19 Exercises (19.1), (19.2)
LB Ch. 16 Propagators & Green's Fns. <i>9 pp • ~1.25h</i>	LB Ch. 20 Scattering Theory <i>7 pp • ~0.5h</i>	LB Ch. 20 Exercise (20.1)
		LB Ch. 7 Exercise (7.1) — bonus

SUN MAY 31 CHECK — MONTH 1 DONE

From memory, write the Feynman rules for LB's Yukawa theory and derive the tree-level cross-section. This closes May — you are caught up on Nuc #1/#2 and ready for June's Standard-Model extension.

JUNE — Standard-Model Deep-Dive

June builds toward the two requested topics — the Higgs mechanism (Week 7) and neutrino oscillations (Week 8) — by laying in gauge invariance, QED, and weak interactions first. Thomson joins as the June companion reader.

WK 5

Gauge Invariance & QED

Jun 1 – Jun 7 • 10 hrs

Goal: bridge LB's toy work into real QED. Four Mann chapters (~60 pp). Ch. 12 (gauge principle) is **the** chapter of the week — it reappears in Weeks 6, 7, 8.

MON	TUE	WED
Mon Jun 1 • 4h • reading	Tue Jun 2 • 4h • reading	Wed Jun 3 • 2h • problems
Mann Ch. 11 Wave Equations <i>14 pp • ~1.5h</i>	Mann Ch. 13 QED (finish) <i>~11 pp • ~2h</i>	Mann Ch. 11 Q2
Mann Ch. 12 Gauge Invariance <i>9 pp • ~1.5h</i>	Mann Ch. 14 Testing QED [skim] <i>19 pp • ~1.5h</i>	Mann Ch. 12 Q1
Mann Ch. 13 QED (start) <i>~8 pp • ~1h</i>		Mann Ch. 13 Q3
		Mann Ch. 14 Q1

END-OF-WEEK CHECK

Explain gauge invariance in one sentence, write the QED Lagrangian, and calculate a tree-level QED scattering amplitude.

WK 6

Weak Interactions

Jun 8 – Jun 14 • 10 hrs

Goal: build the weak-interaction apparatus for Week 7's unification. ~71 pp. End Tue with LB 26 to preview spontaneous symmetry breaking.

MON	TUE	WED
Mon Jun 8 • 4h • reading	Tue Jun 9 • 4h • reading	Wed Jun 10 • 2h • problems
Mann Ch. 19 Beta Decay → Weak <i>22 pp • ~2h</i>	Mann Ch. 21 Charged Weak of Quarks/Leptons <i>24 pp • ~3h</i>	Mann Ch. 19 Q3
Mann Ch. 20 Charged Leptonic Weak <i>16 pp • ~2h</i>	LB Ch. 26 Broken Symmetry (preview) <i>9 pp • ~1h</i>	Mann Ch. 20 Q2
		Mann Ch. 21 Q2
		LB Ch. 26 Exercise (26.1)

END-OF-WEEK CHECK

Sketch the Feynman diagram for neutron β -decay with a W-propagator; explain what the CKM matrix does; state Goldstone's theorem.

WK 7

★ Higgs Mechanism & Electroweak Symmetry Breaking

Jun 15 – Jun 21 • 10 hrs

Capstone topic #1. Densest week — ~100 pp. Strategy: Mann 22 (setup) then attack Mann 23 (derivation). Tue pairs LB 46/47 (QFT voice) with Thomson 17 (LHC/pedagogy voice). **Mann 23 §23.2 is the single most important passage of June.**

MON	TUE	WED
Mon Jun 15 • 4h • reading	Tue Jun 16 • 4h • reading	Wed Jun 17 • 2h • problems
Mann Ch. 22 Electroweak Unification <i>14 pp • ~1.5h</i> Mann Ch. 23 ★ Higgs Mechanism (§§23.1–23.2) <i>~15 pp • ~2.5h</i>	Mann Ch. 23 ★ Higgs Mechanism (§§23.3+) <i>~15 pp • ~1.5h</i> LB Ch. 46 Non-abelian Gauge Theory <i>9 pp • ~0.75h</i> LB Ch. 47 Weinberg-Salam Model <i>11 pp • ~1h</i> Thomson Ch. 17 Higgs (§§17.1, 17.3–17.5) [skim] <i>~18 pp • ~0.75h</i>	Mann Ch. 22 Q2 Mann Ch. 23 ★ Q1 and Q3 LB Ch. 46 Exercise (46.1) LB Ch. 47 Exercise (47.1)

END-OF-WEEK CHECK — HIGGS

Derive the W and Z masses from the Higgs VEV ($v \approx 246$ GeV) and the weak mixing angle. Explain in plain English why the photon stays massless. If both work, you own the Higgs mechanism.

WK 8

★ Neutrino Oscillations & Capstone

Jun 22 – Jun 28 • 10 hrs

Capstone topic #2. ~86 pp. Thomson 13 is the clearest pedagogical treatment; pair it with Mann 25 §§25.1–25.3 for the beyond-SM framing. Mann 24 ties up Week 7's machinery. Thu–Sun buffer: write the 1-page capstone connecting gauge invariance → Higgs → neutrino masses.

MON	TUE	WED
Mon Jun 22 • 4h • reading	Tue Jun 23 • 4h • reading	Wed Jun 24 • 2h • problems
Mann Ch. 24 Testing Electroweak Theory <i>16 pp • ~1.5h</i> Thomson Ch. 13 ★ Neutrino Oscillations (§§13.1–13.5) <i>~22 pp • ~2.5h</i>	Thomson Ch. 13 ★ Neutrino Osc. (§§13.6–13.9) <i>~12 pp • ~1.5h</i> Mann Ch. 25 ★ Beyond SM (§§25.1–25.3) <i>~20 pp • ~2h</i> Thomson §17.8 Addendum: Neutrino Masses <i>3 pp • ~0.5h</i>	Thomson Ch. 13 ★ Problems 13.1, 13.3, 13.5 Mann Ch. 25 ★ Q1 Mann Ch. 24 Q1

END-OF-WEEK CHECK — NEUTRINO OSC.

Derive the two-flavour vacuum oscillation probability $P(\nu_e \rightarrow \nu_\mu)$ from mixing angle θ and Δm^2 . Explain (i) why massless neutrinos can't oscillate; (ii) why the solar-neutrino deficit led to the discovery of oscillations.

CAPSTONE BUFFER — JUN 25–30

Use Thu–Tue as buffer. Write a 1-page capstone connecting gauge invariance → electroweak unification → Higgs mechanism → neutrino masses. If you can draft it without referring back to the books, you own the material.

Progress Tracker (print & tick off)

One row per chapter. Capstone weeks tinted gold (Higgs) and green (neutrino oscillations); ★ rows are the two requested anchor chapters.

Wk	Chapter	Read	Probs	Notes
1	Loveland 8 — Beta Decay	☐	☐	
1	Loveland 9 — Gamma-Ray Decay	☐	☐	
1	Loveland 10 — Nuclear Reactions	☐	☐	
1	Loveland 11 — Fission	☐	☐	
2	Loveland 12 — Nuclear Astrophysics	☐	☐	
2	Loveland 14 — Reactors & Accelerators	☐	☐	
2	Loveland 18 — Radiation Detectors	☐	☐	
3	Mann 3 — Symmetries	☐	☐	
3	Mann 4 — Conservation Laws	☐	☐	
3	Mann 5 — Particle Classification	☐	☐	
3	Mann 6 — Discrete Symmetries	☐	☐	
3	Mann 9 — Scattering	☐	☐	
3	Mann 10 — A Toy Theory	☐	☐	
4	LB 7 — Example Lagrangians	☐	☐	
4	LB 12 — Canonical Quantization Ex.	☐	☐	
4	LB 13 — Many Components / Massive EM	☐	☐	
4	LB 16 — Propagators & Green's Fns.	☐	☐	
4	LB 17 — Propagators & Fields	☐	☐	
4	LB 18 — The S-matrix	☐	☐	
4	LB 19 — Feynman Diagrams	☐	☐	
4	LB 20 — Scattering Theory	☐	☐	
5	Mann 11 — Wave Equations	☐	☐	
5	Mann 12 — Gauge Invariance	☐	☐	
5	Mann 13 — QED	☐	☐	
5	Mann 14 — Testing QED	☐	☐	
6	Mann 19 — Beta Decay → Weak	☐	☐	
6	Mann 20 — Charged Leptonic Weak	☐	☐	

6	Mann 21 — Charged Weak of Quarks/Leptons	☐	☐
6	LB 26 — Broken Symmetry	☐	☐
7	Mann 22 — Electroweak Unification	☐	☐
7	★ Mann 23 — Higgs Mechanism	☐	☐
7	LB 46 — Non-abelian Gauge Theory	☐	☐
7	LB 47 — Weinberg-Salam Model	☐	☐
7	Thomson 17 — Higgs Boson (supp.)	☐	☐
8	Mann 24 — Testing Electroweak	☐	☐
8	★ Thomson 13 — Neutrino Oscillations	☐	☐
8	★ Mann 25 — Beyond SM / Neutrino Osc.	☐	☐
8	Thomson 17.8 — Neutrino Masses	☐	☐

SLIP PLAN

If a Wed problem session runs out of time, prioritize: (1) ★ **starred problems** first, (2) problems from the chapter with the biggest page load, (3) everything else. If a whole week slips, the safe drops are Mann 14, LB 26, LB 46, and Thomson 17 (supplementary). **Never drop** Mann 12, Mann 23, Mann 25, or Thomson 13 — those are the weeks' anchors.